Adaptation

BACKGROUND

habitat (say 'hab-i-tat') – the particular place where a group of living things live

marsupial (say 'marr-soup-ee-al') – a type of small furry mammal, with a pouch for its young

species (say 'spee-seas') – a group of living things that can breed together to produce fertile offspring

Adaptation is the name that scientists give to the way that living things **change** and become **better suited** to their habitats.

Offspring are not identical to either of their parents. Each new generation of a **species** is a tiny bit different from the one before. The **theory of evolution** says that these tiny differences are not noticeable from generation to generation, but over thousands or millions of years these small differences can completely and obviously change the way a plant or animal looks or functions. Adaptation results in new species and new subspecies. In this way it contributes to the **biodiversity**, or biological diversity, of life on Earth.

☞ **1** Explain why you think offspring are not identical to either of their parents.

The process of adaptation happens because of small **variations** within a species. Variations that give some individuals a better chance of survival than other individuals in their species lead to what we call 'survival of the fittest', or **natural selection**. An example of a variation that might give a better chance of survival is a different fur colour, which might make it easier for a small animal to hide from predators.

Because the changed individuals are more likely to survive, they are more likely to breed and produce offspring. Their offspring are more likely to have the changed feature, or **trait**. Over time, this can result in a complete change to all the individuals in a species. The new species trait is called the **adaptive trait**, or the adaptation.

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Adaptation CONT'D

2 Add the missing vowels (a, e, i, o, u) to complete the following words. Hint: All the words have been used on the previous page, in this order.

a) dpttn _____

b) spcs _____

c) vrtns _____

d) srvvl _____

e) prdtrs _____

f) ffsprng _____

g) trt _____

The circumstances within an ecosystem can determine whether a small variation within a species is of benefit or not. For example, within one species of small marsupial in a woodland there are slight variations in fur colour, but most have quite dark fur. Over time, the climate of the forest changes and there is more and more snow. Marsupials that have light-coloured fur are able to escape from predators more often than marsupials with darker fur. So more marsupials with light-coloured fur survive to reproduce, and they reproduce offspring that are more likely to have lighter fur. Over many generations, the species of marsupial in the woodland changes to become white-furred marsupials.

3 Are the following statements about the example adaptation true or false?

a)	Marsupials with light-coloured fur can run faster in the snow.	True	False
b)	Marsupials with light-coloured fur can hide better in the snow.	True	False
c)	Adaptation resulted in two species of marsupials, one light and one dark.	True	False
d)	The change happened quickly but it wasn't permanent.	True	False
e)	The circumstances in the ecosystem determined which individuals were best suited to survive.	True	False

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